

RTVis: Research Trend Visualization Toolkit

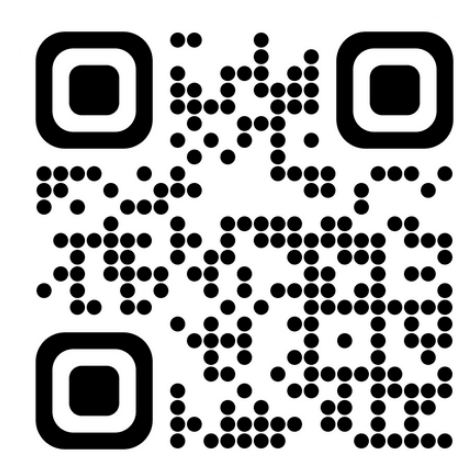
Visualizing the research trend in a single run
Demo: rtvis.design Docs: docs.rtvis.design

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Affiliations

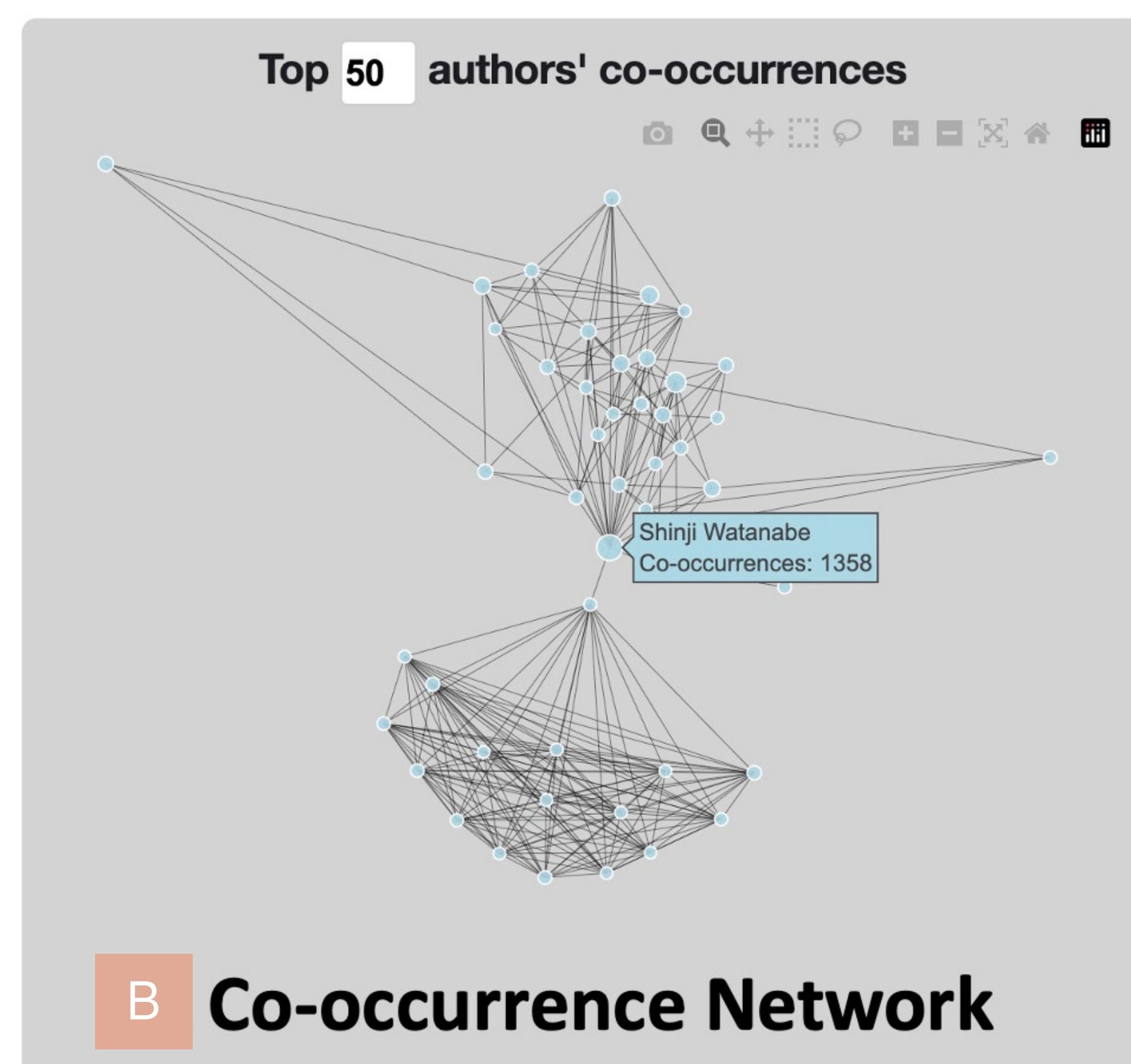
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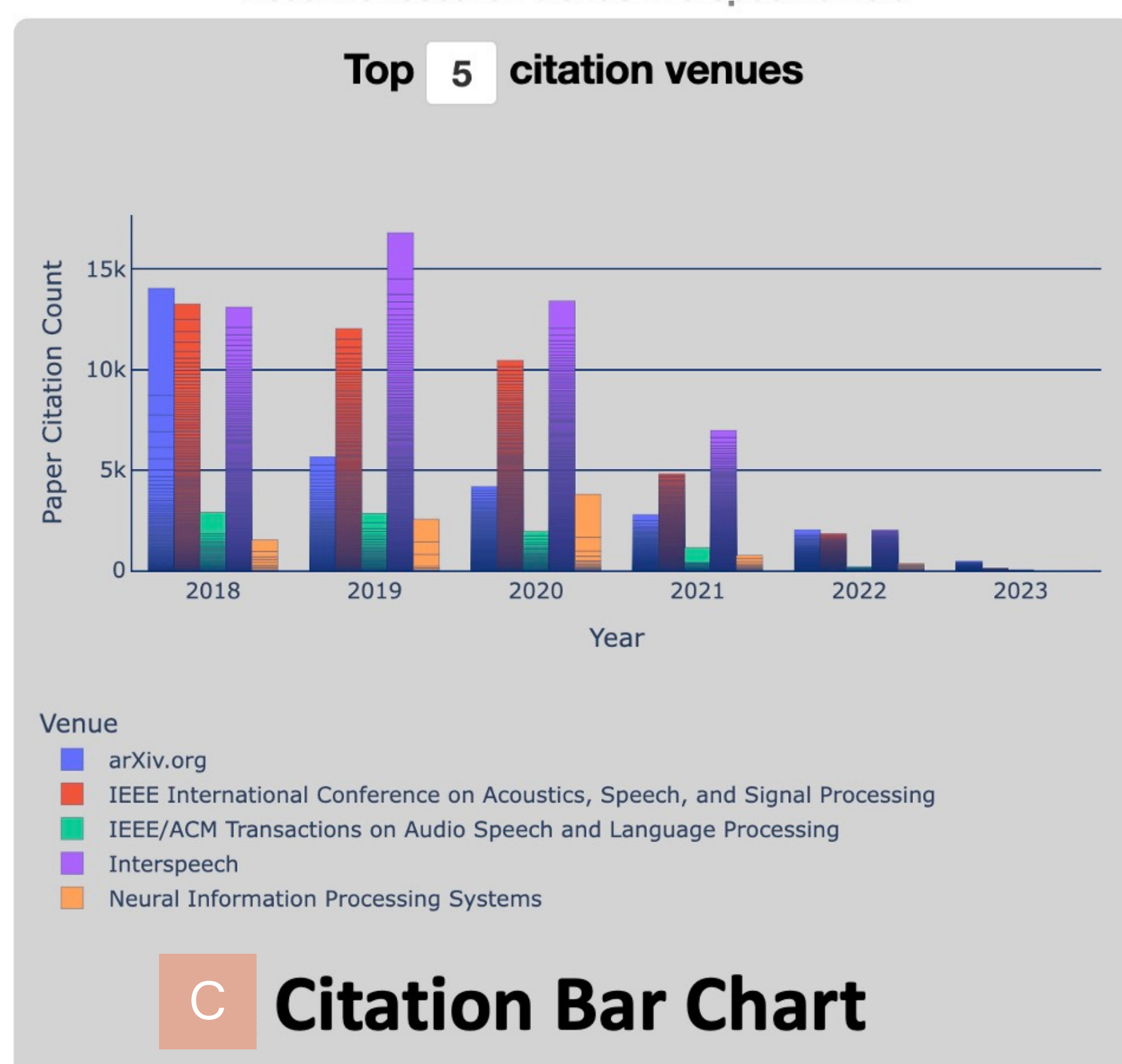
1 Introduction

When embarking on a new research project or entering a novel field, selecting an appropriate topic can be daunting. We introduce the Research Trend Visualization Toolkit (RTVis) to provide real-time insights into research trends. Comprising a field theme river, co-occurrence network, specialized citation bar chart, and word frequency race diagram, RTVis offers a multifaceted view of trends over time, collaboration among authors, venue-specific citation numbers, and prevalent terminology in abstracts. The open-source nature of RTVis ensures accessibility, and it can be easily deployed. Visit our demo or access the code and documentation online.

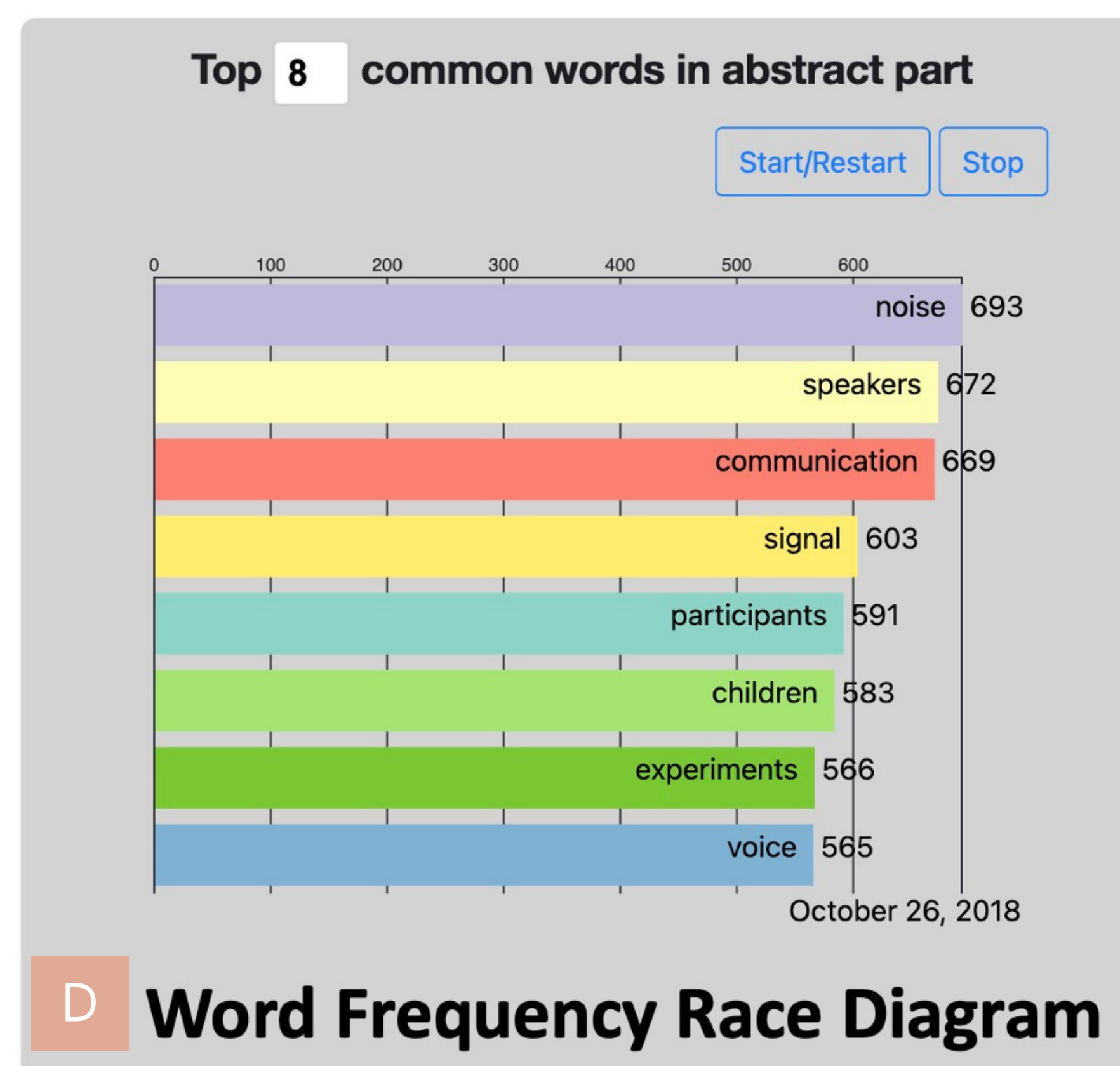
2 Method



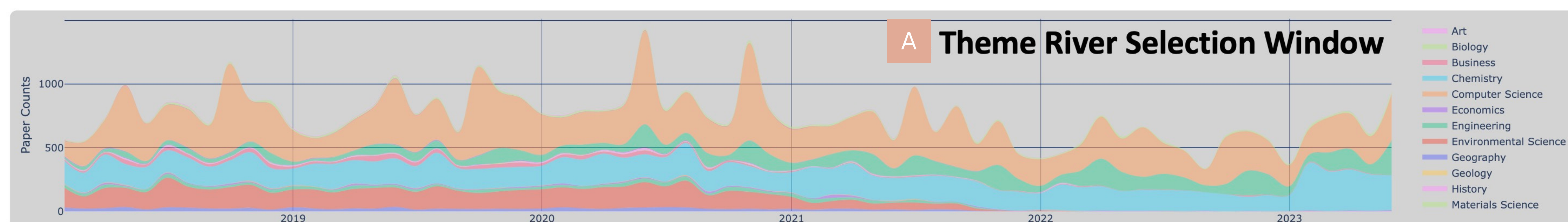
B Co-occurrence Network



C Citation Bar Chart



D Word Frequency Race Diagram



A Theme River Selection Window

A Theme River Selection Window

We developed a theme river visualization that uses horizontal and vertical axes to represent time and research fields, with each river symbolizing a specific field, and its width reflecting the number of papers in that area. This visualization enables researchers to explore and gain insights into research focuses and trends within various fields, capturing their dynamic evolution.

C Citation Bar Chart

The graph presents a comparison of citation counts for papers in various publication venues, with details accessible through interactive elements such as hovering and clicking, enabling users to view individual papers, including their highest cited paper, and directly link to Google Scholar. Additional interactive features include selecting the number of top-cited venues to display, choosing which venues are shown from the legend, zooming in dense areas, and the alignment of paper boxes based on citation count. All above offer users a comprehensive, customizable view of citation trends within a specified time range.

B Co-occurrence Network

The network graph visualizes academic collaborations among authors by representing authors as nodes and collaborations as edges, with the size of each node reflecting the extent of co-authorship, and focuses on the top n authors based on co-authorship frequencies. This method showcases the algorithm's robustness, facilitates a comprehensive analysis of the collaboration network and identifies influential authors and their interconnected relationships.

D Word Frequency Race Diagram

The diagram dynamically animates bars representing frequently occurring words in abstracts, with lengths proportional to frequency, illustrating fluctuations in word usage across different abstracts, enabling the identification of key topics and themes, and allowing comparisons across time periods or various research fields. Additional features such as an input box for customization of the number of bars provide researchers with valuable insights into the evolution and variations of word prominence within specific subjects, as well as the dominant interests and directions of a particular field.

3 Evaluation

The RTVis toolkit skillfully amalgamates a spectrum of proven visualization techniques to elucidate various facets of academic research. This adaptability widens its applicability and prospective impact across multiple disciplines. For example, individuals at the outset of their research in fields like speech signal processing can benefit from RTVis to understand prevailing research directions. Through the theme river visualization, our sample data showed that most papers emanate from the computer science and medical sectors. Interestingly, the number of psychology-related papers exhibited a consistent decline, almost reaching a null point by the end of 2021.

Our toolkit employs a suite of visualizations for an encompassing academic assessment. The co-occurrence network, for instance, spotlights influential authors such as Shinji Watanabe and Brian MacWhinney, who serve as hubs connecting smaller research clusters. Citation bar charts reveal high-impact venues like Arxiv, Interspeech, and ICASSP, where papers like Word2vec and Comformer are prominently cited. The frequency race diagram, based on our dataset, displayed eight bars; the animation made it evident that terms related to computer science were predominant from 2018 to 2023.

By integrating all these visualizations, researchers could readily discern 1) the prevailing research topics, 2) the influential authors in the field, and 3) the most suitable conferences for publication. We posited that through our proposed visualizations, researchers were endowed with sufficient information to explore their selected fields.

4 Conclusion

We present RTVis, an open-source toolkit for understanding research trends in a specific period. RTVis allows users to visualize collaborations among authors, explore significant publication venues, analyze prevalent topics, and discover prominent research fields—all within a single web page. It supports datasets collected by users or sourced from Semantic Scholar and can be easily deployed on local machines or servers with a single run. While our toolkit has demonstrated promising results, there is still room for improvement. In the data processing stage, the accuracy and efficiency heavily rely on the quality of the input data, while data collection and cleaning are time-consuming. To enhance accuracy and efficiency, we plan to implement advanced algorithms that explore machine-learning approaches to automate the data-cleaning process and improve the overall reliability of the toolkit. The updating process may take longer as the system requires recalculating all components during live updates. We aim to optimize the visualization pipeline by pre-computing and caching frequently accessed calculations or visualizations to address this. This will substantially reduce the processing time required for live updates and rendering on low-computational-resource devices. Additionally, we will proactively seek user feedback through surveys and conduct user studies to identify any issues or areas for improvement. By addressing these challenges and incorporating user insights, we are confident that our toolkit will evolve to offer improved and user-friendly functionalities.

References

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